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Internet of Things Enabled Power Theft Detection and Smart Meter Monitoring System

M J Jeffin, Madhu G M, Akshayata Rao, Gurpreet Singh and C. Vyjayanthi

***Abstract*—Power theft, at low voltage distribution end is a concerning issue as the distribution companies lose billions of revenue annually. With the advent of smart grid technologies, smart meters with Information Communication Technology (ICT) can provide a solution for detecting and alerting the power theft. This paper presents the application of Internet of Things (IoT) in power theft detection and real-time smart meter monitoring. Linear Regression method is used for detecting power theft by continuously monitoring the consumer and distribution end smart meters data. Android applications are developed for monitoring consumption & billing information of consumers and alerting the authorities in the event of theft. The presented system is capable of detecting power theft due to meter bypass, meter tampering and direct line hooking. As an additional feature, direct control of smart meters from distribution authorities is implemented for providing access/denial of power supply for an individual consumer. A prototype circuit is developed using ATmega328P micro-controller with NodeMCU as a WiFi module, for validating the presented system.**

***Index Terms*—Internet of Things (IoT), Information Communication Technology (ICT), Linear Regression, Power Theft, Smart Meter Monitoring.**

I. Introduction

POWER loss is one of the important issues that has to be taken care in the distribution network. It can be addressed as the difference in power, between the generation and metered consumption. This power loss is classified into technical losses and non-technical losses. Majority of the power loss are non-technical losses and are due to power theft [1]. Power theft can be defined as the illegal or un-metered utilization of electricity from distribution utilities. The distribution utilities incur huge financial losses due to this power theft. It has been estimated that around 6-10 billion INR of revenue will be lost in India because of the theft every year [2].

Consumers commit power theft in various ways which include meter bypassing, direct line hooking, meter tampering etc. [3]. A very common way of bypassing the meter include, connecting the supply wire directly to the distribution network in parallel to the meter [4]. The other way to access electricity is by tampering the meter. This type of theft is generally practised in the village and suburban areas where, insertion of foreign objects, magnetic interfering materials into meters or shortening meter terminals are done so that meter is not operated [5]. Apart from these, consumers also commit direct line hooking on low voltage overhead lines to access the electricity. Hence power theft is a critical issue for the distribution utilities as it directly reflects on the revenue of the utility. Substantial research has been carried out on power theft detection algorithms. [6]-[8] presents the power theft detection methods using consumer load profile analysis. But these methods have a limitation that they cannot identify complete bypass and line hooking type of thefts. [9] presented a method to identify power theft using temperature based predictive models which uses meter data and distribution transformer information. In [10], authors applied state estimation for power theft detection where, a privacy preserving theft detection algorithm has been developed by applying Kalman filtering approach. [11][12] presents the linear regression method for theft detection. however, it mainly concentrates on mathematical design of theft detection algorithm and less on real-time implementation. Apart from these, [13] presented a power line communication-based approach for theft detection. In [14] the authors designed an inspection device to detect power theft based on the magnetic oscillations of the distribution line. Various approaches have been presented in the literature for theft detection. However, these methods are more complex, computationally difficult and the utility has to invest in building dedicated infrastructure at consumer premises which is not feasible. Moreover, majority of methods concentrate only on detecting the theft and there is less research explored in building communication technologies to alert the authorities about the theft.

With the advent of smart grids, the conventional meters are replaced with the smart meters and these smart meters are capable for upgrading with communication technologies so

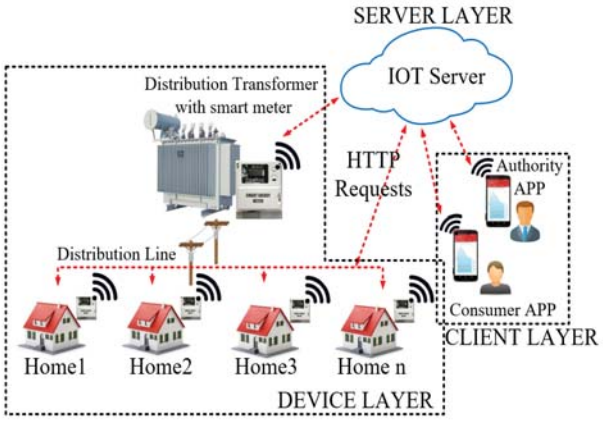


Fig. 1. Three Layered Architecture of Smart Meter based Power Theft detection System.

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that they can interact with central hub/server for real time monitoring. Internet of Things (IoT) provides the advanced communication technologies through which the consumer utilization information can be sent to the central server. By utilizing each consumers information along with distribution transformer information, any type of power theft can be detected and alert the concerned authorities at distribution transformer level. Few research works related to application of IoT in smart metering and theft detection have been recorded in the literature. [15] presents an IoT based smart meter which has advanced features like meter tampering alert. But the presented system does not possess theft identification. The authors in [16], presented a smart meter which has the feature to send all metered data to the server. This device uses two controllers one for communication and other for computation and the presented meter is not cost effective. Hence, it is identified that, there is a need for developing a system which would detect power thefts as well as monitor individual consumer utilization. In this regard this paper presents an IoT based power theft detection and smart meter monitoring system which uses linear regression-based approach for detecting the power theft. The proposed system is made to be low computational and cost effective so that its practical implementation is feasible.

The rest of the paper is organized as follows: Section II gives the architecture and working of the presented system followed by section III gives the prototype hardware implementation. In section IV the results are presented and the paper is concluded in section V followed by the references.

II. Proposed System Architecture & Working

Fig. 1 presents the client server model based architecture of the proposed system. It consists of three layers namely device layer, client layer and server layer. The smart meters at different nodes form a device layer. The android application developed for consumers and authorities forms a client layer and the cloud server forms an IoT server layer.

These layers interact with each other through internet using Application Programming Interfaces (APIs) and HTTP requests. The server uses four kinds of requests for communicating, updating, retrieving and deleting the data. They include GET request for retrieving the data from the server, PUT request for updating a value in the database of a server, POST request for continuous feeding of real time value to server database and DELETE request for removing an entry from the database.

All the devices, in device layer and client layer communicate directly with the server. The real-time energy information is sent from the smart meters to server and the server process these information to identify the theft. This information about the theft is sent to clients using APIs.

A. Design of IoT enabled Smart Meter with Tampering Detection circuit

Smart meters are important in proper operation of the presented system. A design of IoT enabled smart meter with

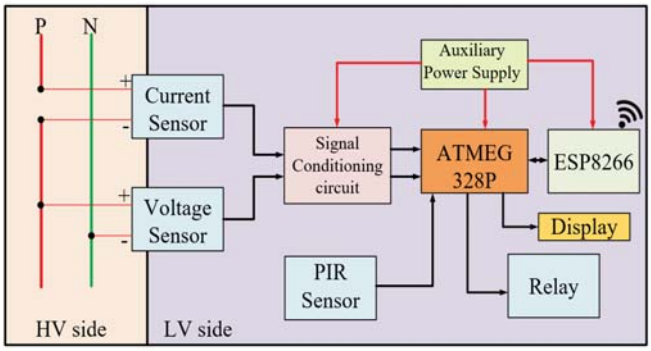


Fig. 2. Block diagram of IoT based Smart Meter.

Tampering detection circuit is presented in Fig. 2. This smart meter has the capability to directly interact with the IoT server and have the feature to turn ON/OFF the supply to a consumer based on the commands from the server or authority. The readings measured at every interval is sent to the server through HTTP requests and these readings along with billing information can be accessed by the consumer using android application. A tampering detection circuit is developed which continuously monitors for meter tampering. On detection of the meter tampering, the smart meter automatically sends an alert message to the concerned authority and stops the supply to consumer.

B. Linear Regression based Power Theft Detection

Considering a distribution transformer supplying "n" homes as shown in Fig. 1. where, each consumer is equipped with smart meter which monitor the consumption of utilized energy individually. A smart meter is installed near the distribution transformer which cumulatively measures the power outflow from the transformer to all "n" consumers.

Let, the power consumed by each home be P(n) and Power sent from the transformer be C(k).

Where "n"corresponds to consumer number and "k" is the transformer number under a feeder.

Then, the power consumed over a duration "t" can be given as,

$$C(k)_t = P(1)+P(2).....+P(n)+TL+EM+PT \quad (1)$$

Where, TL is the technical losses occur in the distribution network, EM is the measurement errors occurred in meters. And PT is the loss occurred due to power theft.

Then (1) can be rewritten as,

$$C(k)_t = \sum_{i=0}^{n} P(n)_t + TL + EM + PT \quad (2)$$

Considering the TL and EM to be negligible, the power theft can be given as,

$$PT = C(k)_t - \sum_{i=0}^{n} P(n)_t \quad (3)$$

Ideally, If PT > 0 then there is a theft.

For practical considerations and to include the technical loss

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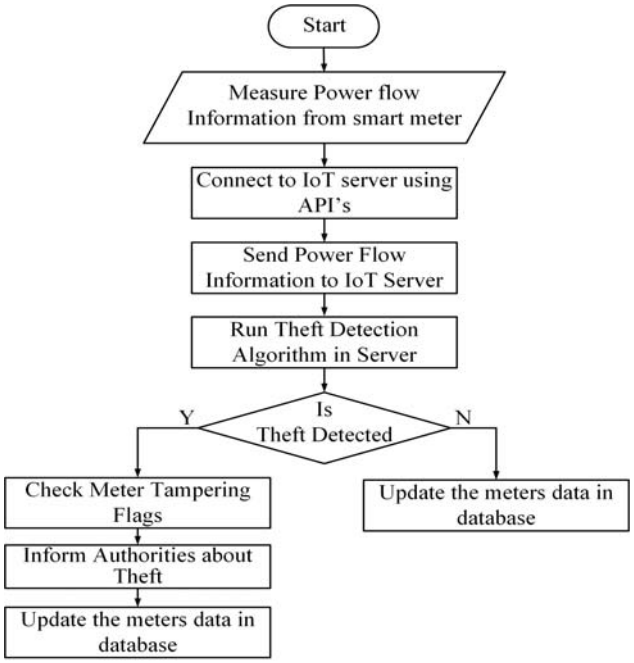


Fig. 3. Work Flow of Theft Detection and Smart Meter monitoring.

factor, PT should be greater than a threshold Value. This threshold value will be chosen suitably based on the system ratings and amount of power flow.

C. Overall Work flow of the system

The proposed system works according to the flowchart given in Fig. 3. Initially, all the smart meters at individual consumer premises and the smart meter equipped near the distribution transformer reads the power flow parameters and sends to the server. In server, the algorithm compares the power flow information of transformer with cumulative power flow of all consumer meters to arrive at theft happening. If there was any theft happened then the server checks for any meter tampering/bypass to identify the theft due to tampering/bypass. If there is any tampering/bypass, that corresponding consumer information will be passed to the authorities. If all the meters are not tampered/bypassed, then the theft occurred will be due to direct line hooking. Hence server sends this information to authorities to check the illegal direct access.
Apart from this, the consumer utilization data will be stored in the database for billing. Consumer can access his utilization and billing information at any time using the android application.

III. System Implementation

Implementation of the presented system is divided into two parts viz software design and hardware development. In software development, Message Queuing Telemetry Transport (MQTT) based server is created and an android application is developed. In hardware development a low cost IoT enabled smart meter with tampering protection is developed.

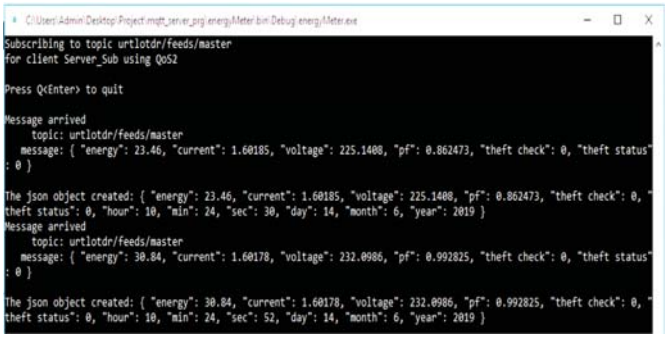


Fig. 4. Snapshot of Developed MQTT based IoT server.

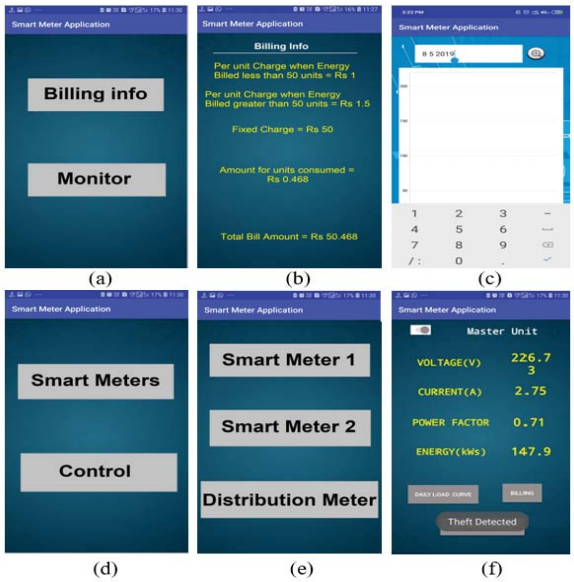


Fig. 5. Snapshots of Developed Android Application. a)-c) Consumer App Activities d)-f) Authority App Activities.

A. Software Design

1) IOT Server: A MQTT protocol-based server is developed using publish subscribe architecture for theft detection and smart meter data monitoring. Eclipse and C language is used to create the publish and subscribe commands. Different databases are used for storing the meter data of consumers. Fig. 4 shows the snapshot of developed server.

2) Android Application: An android application is developed using android studio 2.3 IDE (Integrated Development Environment) for consumers and authorities. API 23 (Marshmallow) software development toolkit (SDK) is used in developing the application. The consumer app consists of billing activity to check the billing information and real time energy monitoring activity for seeing the present and past energy consumption information. Similarly, the app developed for authorities consists of access to all smart meter data and turn ON/OFF feature in a control and monitor activity. These activities are designed used XML language and the back-end operation is coded using derived Java for android. Fig. 5 presents the photographs of developed activities. Algorithms 1 and 2 gives the activity life cycle of the developed applications.

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Algorithm 1: Activity life cycle for Consumer App

1. Splash Activity → Delay
2. Main Activity,
   a) Initialize onclick listener for Selection option between billing and monitoring button
   b) On click, → move to corresponding activity
   c) Update memory
3. Billing Info Activity,
   a) Connect to WiFi → return acknowledgement.
   b) Connect to server using API.
   c) Fetch Consumer billing info.
   d) Display results.
   e) On back button pressed →return to previous activity state from memory.
4. Monitor Activity,
   a) Connect to WIFI → return acknowledgement.
   b) Connect to server using API.
   c) Fetch Consumer utilization info of corresponding date.
   d) Display results in graph.
   e) On back button pressed → return to previous activity state from memory.

Algorithm 2: Activity life cycle for Authority App

1. Splash Activity → Delay
2. Main Activity,
   a) Initialize onclick listener for Selection option between smart meters and control button
   b) On click, → move to corresponding activity
   c) Update memory
3. Smart Meter Activity,
   a) Initialize onclick listener for Selection option between smart meters.
   b) Onclick, Connect to WIFI → Connect to server using API.
   c) Fetch smart meter info.
   d) Display results.
   e) On back button pressed →return to previous activity state from memory.
4. Control Activity,
   a) Connect to WIFI →Connect to server using API.
   b) Send commands to either turn ON / OFF the smart meter return → response.
   c) On back button pressed → return to previous activity state from memory

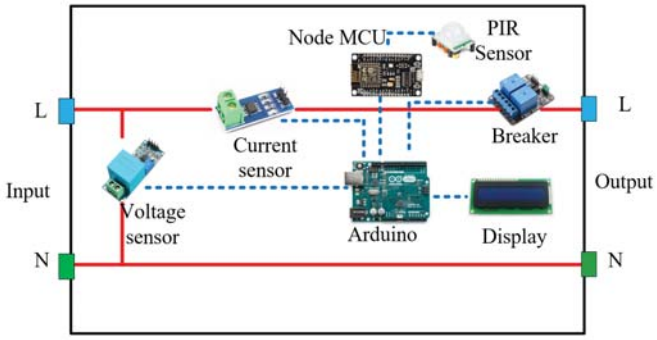


Fig. 6. Circuit Diagram of the Smart Meter.

B. Hardware Design

Fig. 6 presents the circuit diagram of the smart meter. The smart meter is designed to have communication features so that it can connect to IoT server. The presented meter uses Arduino/ATMEG 328P as main controller and Node MCU for adding WiFi feature. Voltage and current sensors are used to read the corresponding voltage and current information and generate equivalent analog signals. These analog signals are passed to the Arduino microcontroller where active power (P), reactive power (Q), power utilized (KWH) and Power Factor (PF) is calculated.

These values are updated to server through Node MCU. A Passive Infrared Sensor (PIR) sensor is used to detect the tampering. If there was any tampering, the output of PIR sensor will be HIGH. Based on the output of PIR sensor an alert message is passed. A circuit breaker is used to control the access/denial of power to a consumer as shown in Fig.6.

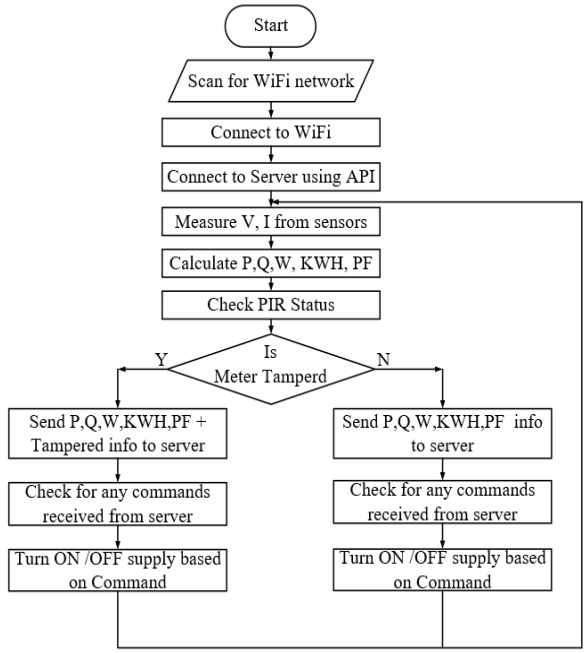


Fig. 7. Flowchart of the Smart Meter Working.

The breaker is connected in NC (Normally Closed) manner and whenever server sends the command to turn OFF then the input of the breaker is made HIGH, in turn breaking the circuit. Fig.7 shows the overall working of the presented smart meter.

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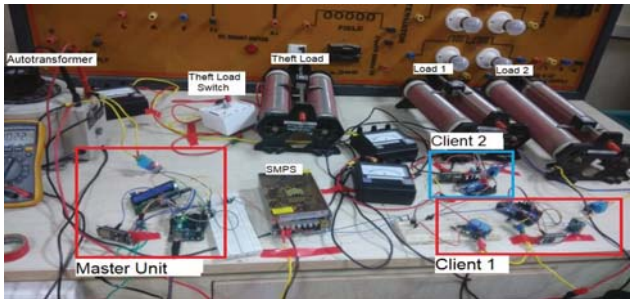


Fig. 8. Prototype system Developed with two clients and a master unit.

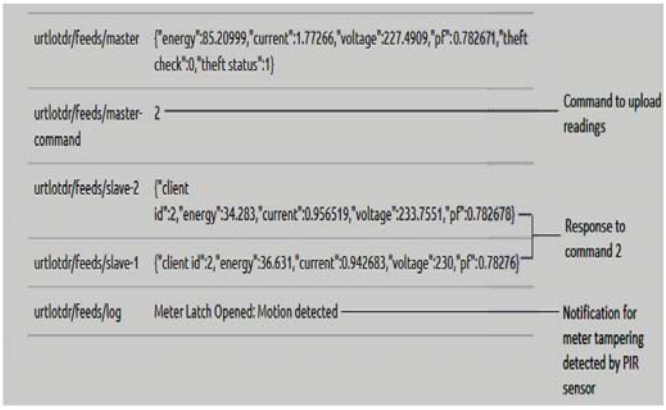


Fig. 9. Server-Side Event Log during the Meter Tampering.

C. Implementation

A prototype system consisting of two client meters (depicting consumers) and a master unit meter (depicting distribution transformer meter) is developed for validating the presented system. Fig. 8 shows the snapshot of the implemented hardware setup. An MQTT server is created using python and eclipse Paho in a local computer, acting as an IoT server and smart meters are coded with API's to access this server.

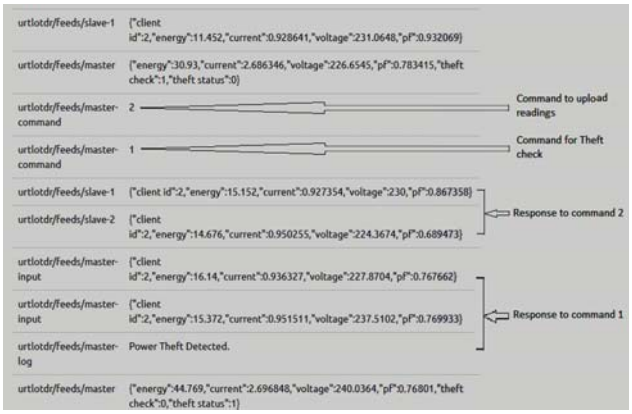
The two client meters and the master unit meter sends the metering data to the IoT server using publish subscribe commands. This metering information is used to detect the theft in server, in turn if there was a theft, concerned authority will receive the alert message. Apart from this, the server stores the metering information in separate database created. A separate API is created to access the metering information of a particular consumer for a particular date and time and plot the graph out of it.

IV. Results & Discussion

Different theft case scenarios are mimicked to check the working of the prototype system like, meter tampering, direct line hooking and meter bypass. This section gives the details of the test conditions carried out.

A. Theft due to Tampering

A scenario is created to tamper the smart meter of client 1 by making the PIR sensor to give motion detect output. The smart meter reads the sensor value and sends the information to server detecting the meter tampering event. Fig. 9 shows the server-side data received from the client 1 smart meter.



(a)



(b)

Fig. 10. Theft Detection due to Direct Line Hooking and Meter Bypass. a) Server Event Log b) Authority App Receiving Notification.

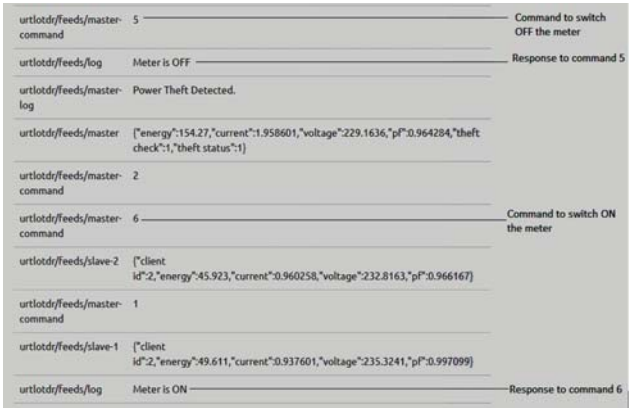


Fig. 11. Server Sending Stop command to Meter in the Event of Meter Tampering.

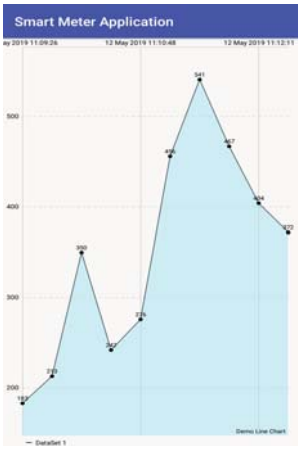
B. Direct line hooking and meter bypass

An additional theft load along with bypassing client 2 meter has been done to mimic the theft. The server based upon the data received from the smart meters detected the theft and the message is sent to the authority app regarding the theft. Fig. 10 shows the event log and app receiving theft alert.

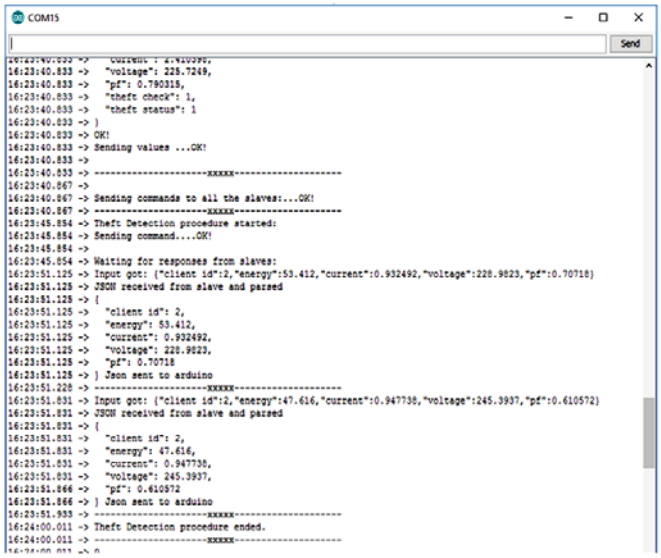
C. Controlling smart meter from server

During the event of meter tampering the server is coded to send command to stop the supply to the consumer. Hence when the tampering is detected server sent a command to stop the supply

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(a)



(b)

Fig. 12. Real Time Energy Utilization Information a) From Consumer App b) Event Log in Smart Meter.

and acknowledgement is received from the smart meter. Fig. 11 gives the event log happened during this condition.

D. Smart Meter Monitoring

The presented system is expected to monitor the individual client metering information and send this information to client app when requested. Fig. 12 presents the client app reading the real time energy consumption information with time stamp and the smart meter sending the data to the server.

V. Conclusion

This paper presents a framework for IoT based real time power theft detection and smart meter monitoring system. A linear regression-based approach is presented to detect the power theft. A new IoT based architecture is used for implementing communication infrastructure for the theft identification and alerting system. Android applications are developed for easy monitoring of energy utilization/billing information and alerting the theft for authorities. A prototype system is implemented and various test scenarios are considered to validate the proposed system. The future work for the presented system lies in improving the theft detection algorithm to use past history of each consumers to detect the theft at each smart meter level.

Acknowledgement

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